

6 (a) (i) The direction of induced e.m.f. produces effects that oppose the change in magnetic flux **B1**

(ii) As the current in the solenoid is switched on, the magnetic field in solenoid is increases. **B1**

This causes an increase in magnetic flux linkage through the small coil.

B1

By Lenz's law, the magnetic field in the coil point upwards to oppose the increase in flux linkage. **B1**

(iii) **B1**



(b) e.m.f. = $N\Delta\phi / \Delta t$ **C1**

$$= (75 \times 1.4 \times 10^{-3} \times 7.0 \times 10^{-4}) / 0.12 \quad \text{C1}$$

$$= 6.1 \times 10^{-4} \text{ V} \quad \text{A1}$$

7 (a) (i) $Q = (M_1 - M_2 - M_3)c^2$